

BASIC PRINCIPLES OF MEDICINE 1:
FOUNDATIONS TO MEDICINE

LECTURE 21

Second messenger systems

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Objectives: Second Messenger Systems

SOM.MK.I.BPM1.1.FTM.3.BCHM.0127	Explain how cholera toxin modifies Gs and how pertussis toxin modifies Gi. Predict the effects of these modifications in the respective cells.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0128	Discuss the roles of cAMP, IP3, DAG and Calcium as mediators of signal transduction in different cell types.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0129	Review the mechanism of action of phosphodiesterase inhibitors.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0130	Describe activation of adenylate cyclase by Gs alpha and phospholipase C by Gq-alpha subunits.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0131	Describe the formation and biological role of cGMP. Indicate significance of guanylate cyclase.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0132	Identify mechanism of action of nitric oxide (nitroglycerin) and its role in smooth muscle relaxation and vasodilatation.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0133	Compare the cellular activation of protein kinases A, C and G.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0134	Distinguish signaling mechanisms associated with alpha-1, alpha-2 and beta-adrenergic receptors, glucagon receptors and insulin receptors.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0135	Predict the changes when an agonist or antagonist of a specific signaling pathway is administered.

Recommended reading

- Lippincott's Biochemistry 9th Edition 
<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
 - Chapter 8: Introduction to Metabolism and Glycolysis) - Section: Metabolism Regulation
- [Lippincott® Illustrated Reviews: Cell and Molecular Biology, 3e | Premium Basic Sciences | Health Library](#); Chapters 17, 18, 19.
- Lippincott Illustrated Reviews: Biochemistry, 9e: Chapter 8 (Introduction to Metabolism and Glycolysis) - Metabolism Regulation: [Introduction to Metabolism and Glycolysis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Practice Questions: [8.6](#)
- Reference: Marks' Basic Medical Biochemistry: A Clinical Approach, 6e: Chapter 10. [Cell Structure and Signaling by Chemical Messengers | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library](#)

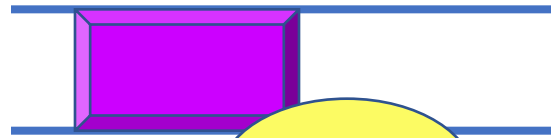
Adrenergic receptors
generate different
second messengers

Epinephrine or
Norepinephrine

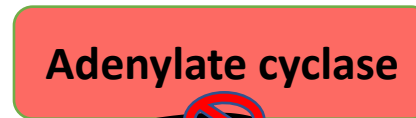
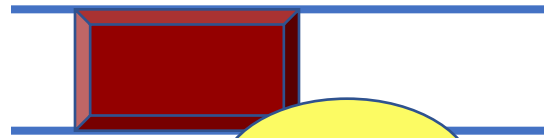
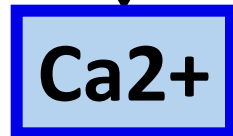
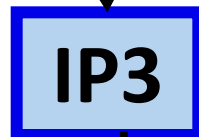
α -1 adrenergic receptor

α -2 adrenergic receptor

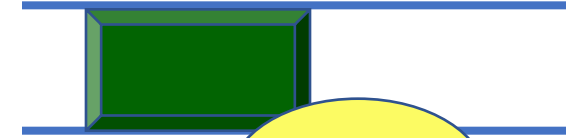
β -adrenergic receptor



Phosphatidylinositol
bisphosphate (PIP2)



ATP



ATP



Second messenger systems - cAMP

Stimulus

Epinephrine/Norepinephrine

Glucagon

Receptors

β -adrenergic receptor

Glucagon receptor

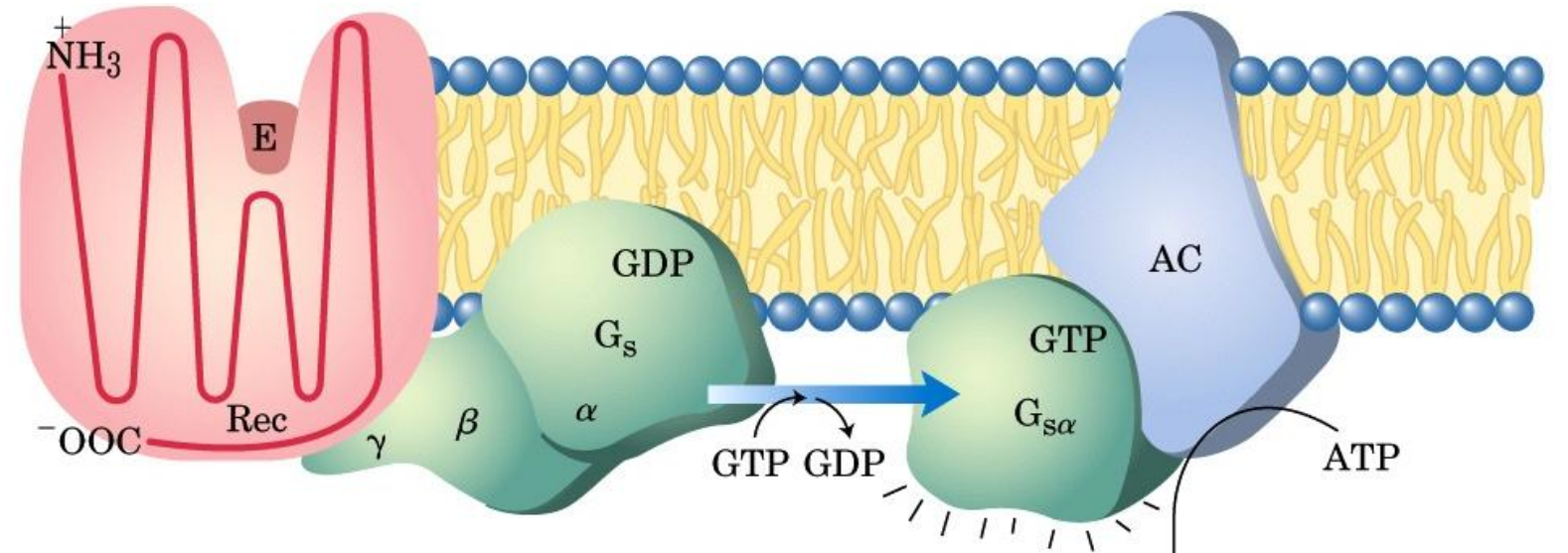
Activates Adenylate cyclase

Increases intracellular cAMP

Activates Protein kinase A

Intracellular effects

1. Epinephrine binds to specific **beta-adrenergic** receptor



② The occupied receptor causes replacement of the GDP bound to G_s by GTP, activating G_s .

③ G_s (α subunit) moves to adenylyl cyclase and activates it.

④ Adenylyl cyclase catalyzes the formation of cAMP.

⑤ PKA is activated by cAMP.

⑥ Phosphorylation of cellular proteins by PKA causes the cellular response to epinephrine.

⑦ cAMP is degraded, reversing the activation of PKA.

Second messenger systems - cAMP

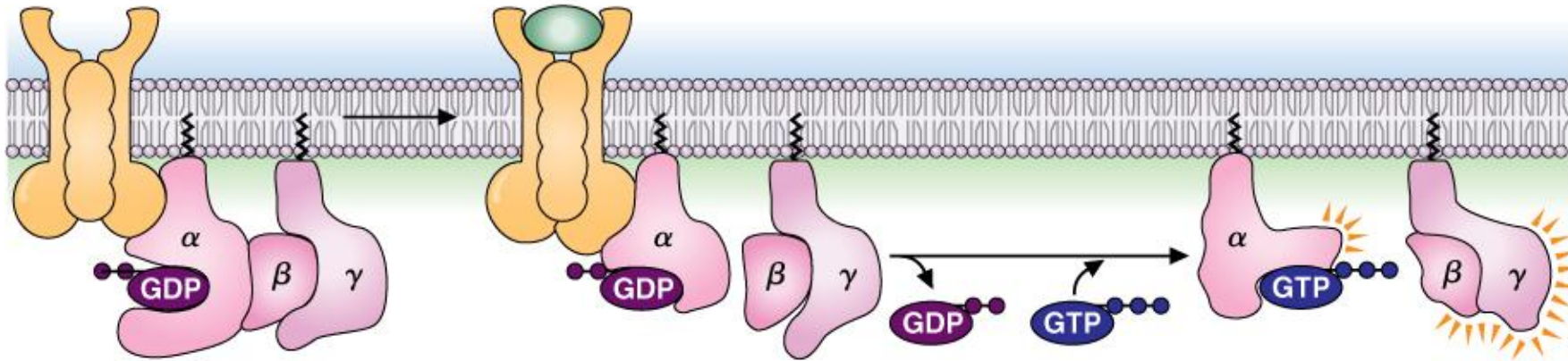
Changes in G-protein on formation of hormone-receptor complex

SOM.MK.I.BPM1.1.FTM.3.BCHM.012

6

1. Hormone binds to receptor

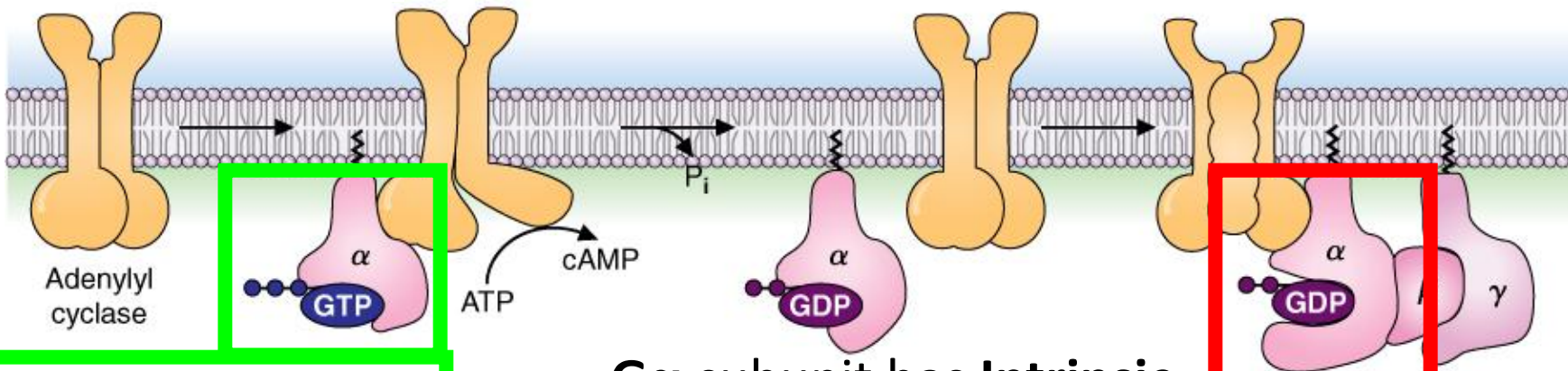
2. Exchange of GDP for GTP on $G\alpha$. Dissociation of $\beta\gamma$ -subunits



3. Activation of Adenylate cyclase by $G\alpha$ -GTP

4. $G\alpha$ has intrinsic GTPase and dissociates from adenylate cyclase

5. $G\alpha$ s-GDP reassociates with $\beta\gamma$ -subunits



$G\alpha$ -GTP is active

$G\alpha$ subunit has Intrinsic GTPase activity

$G\alpha$ -GDP is inactive

**Epinephrine or
Norepinephrine**

Glucagon

β -adrenergic receptor

Glucagon
receptor

Gs α

Beta-adrenergic
receptor and glucagon
generate cAMP

**Gs α can be activated
by many hormones**

Adenylate cyclase

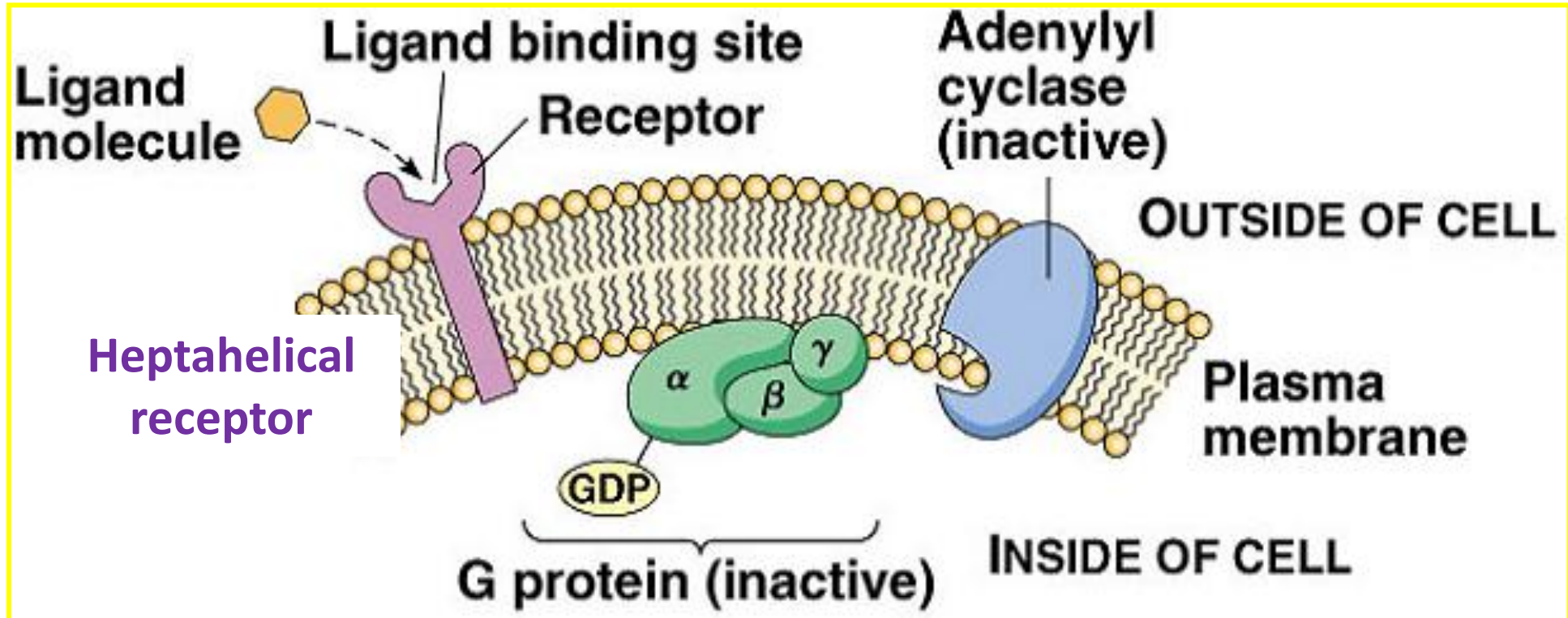
ATP

↑cAMP

Activates Protein kinase A

Steps in Adenylate cyclase activation

1. Hormone **binds** to G-protein coupled receptor

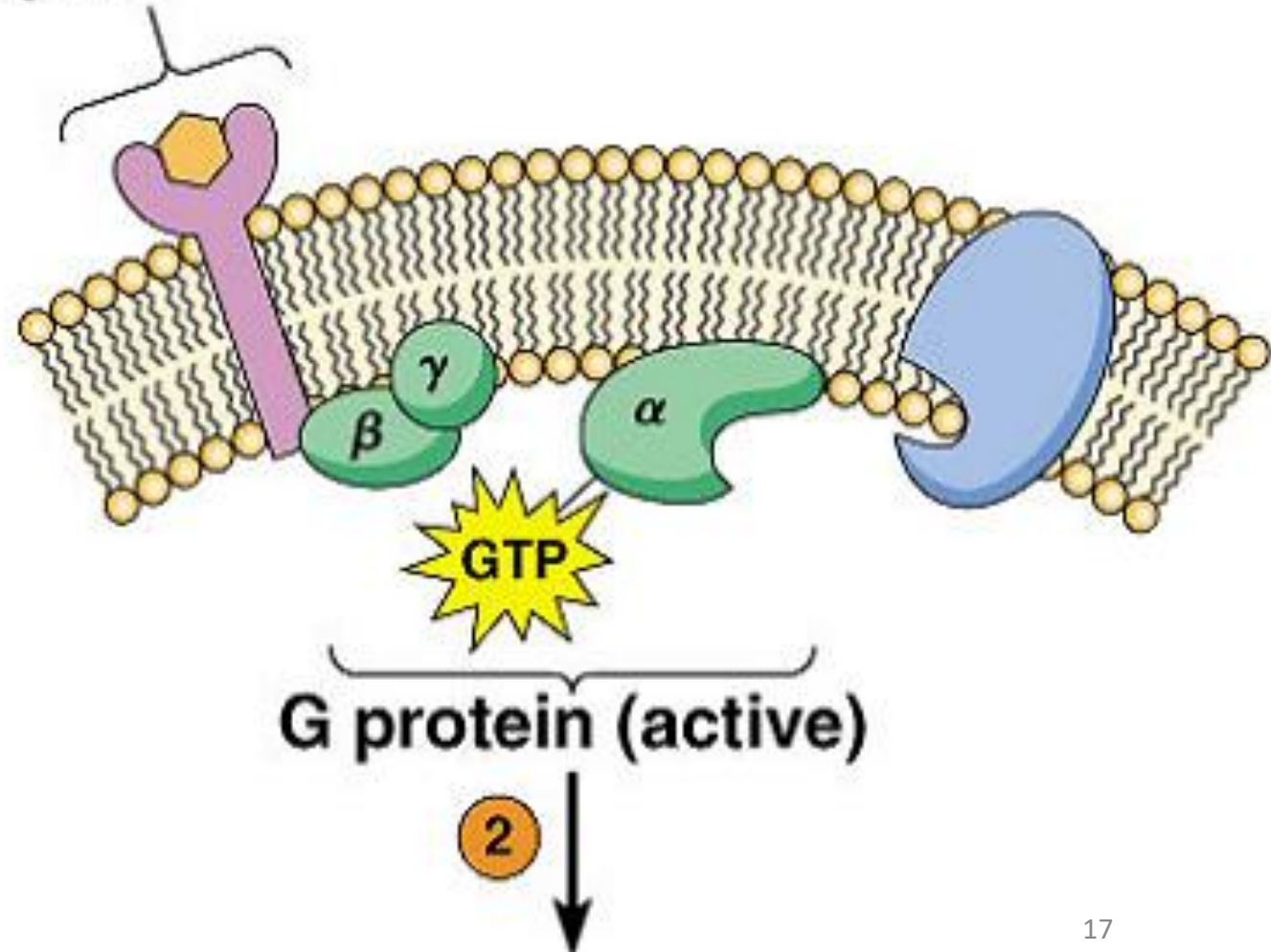


The inactive G protein is bound to GDP

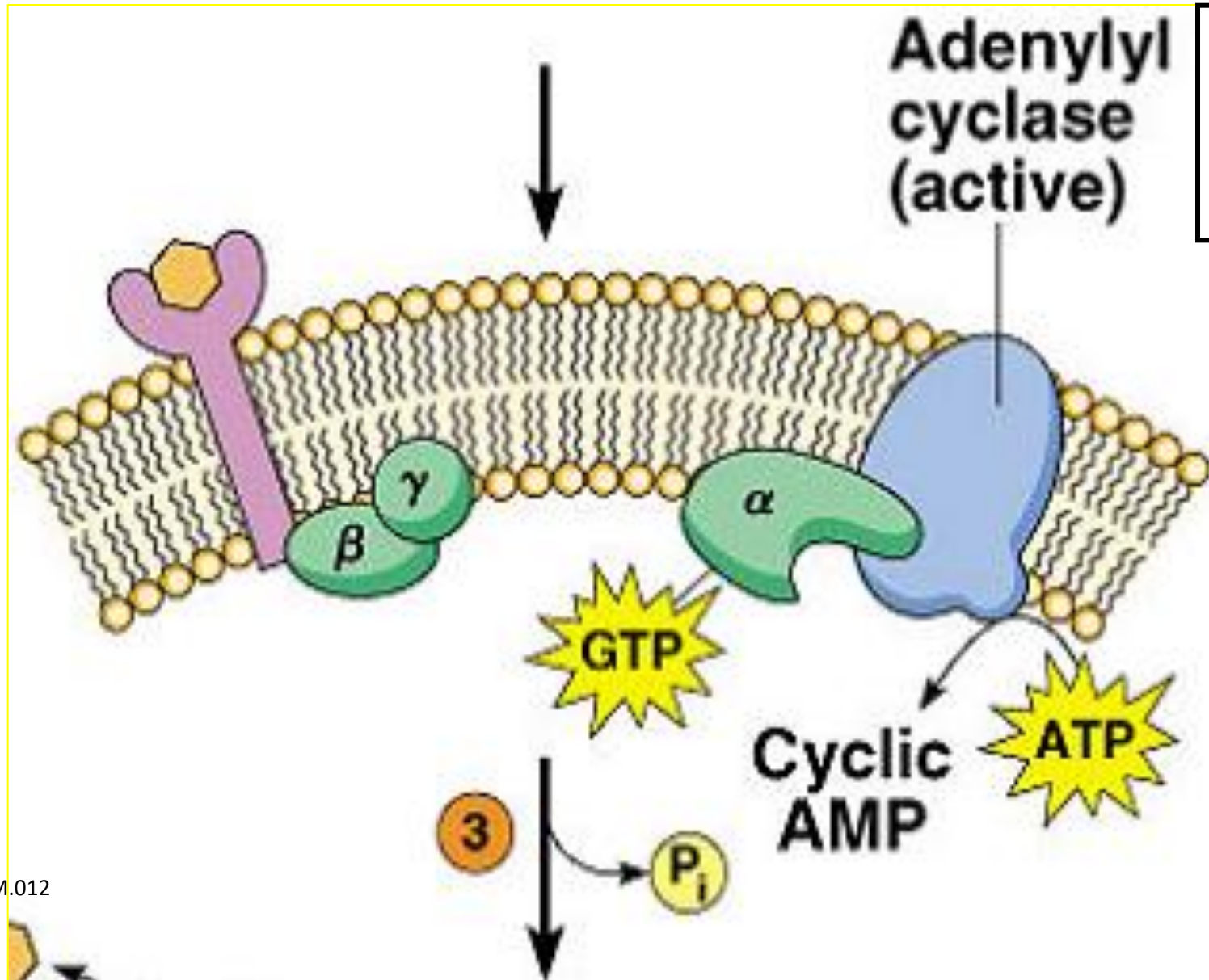
Steps in Adenylate cyclase activation

2. Activated receptor interacts with **G-protein** causing **G α subunit** to release GDP and bind **GTP**; **G α subunit-GTP** bound and **$\beta\gamma$ dimer** dissociate

Ligand-receptor complex



Steps in Adenylate cyclase activation



3. GTP bound $G\alpha$ subunit activates **adenylate cyclase**

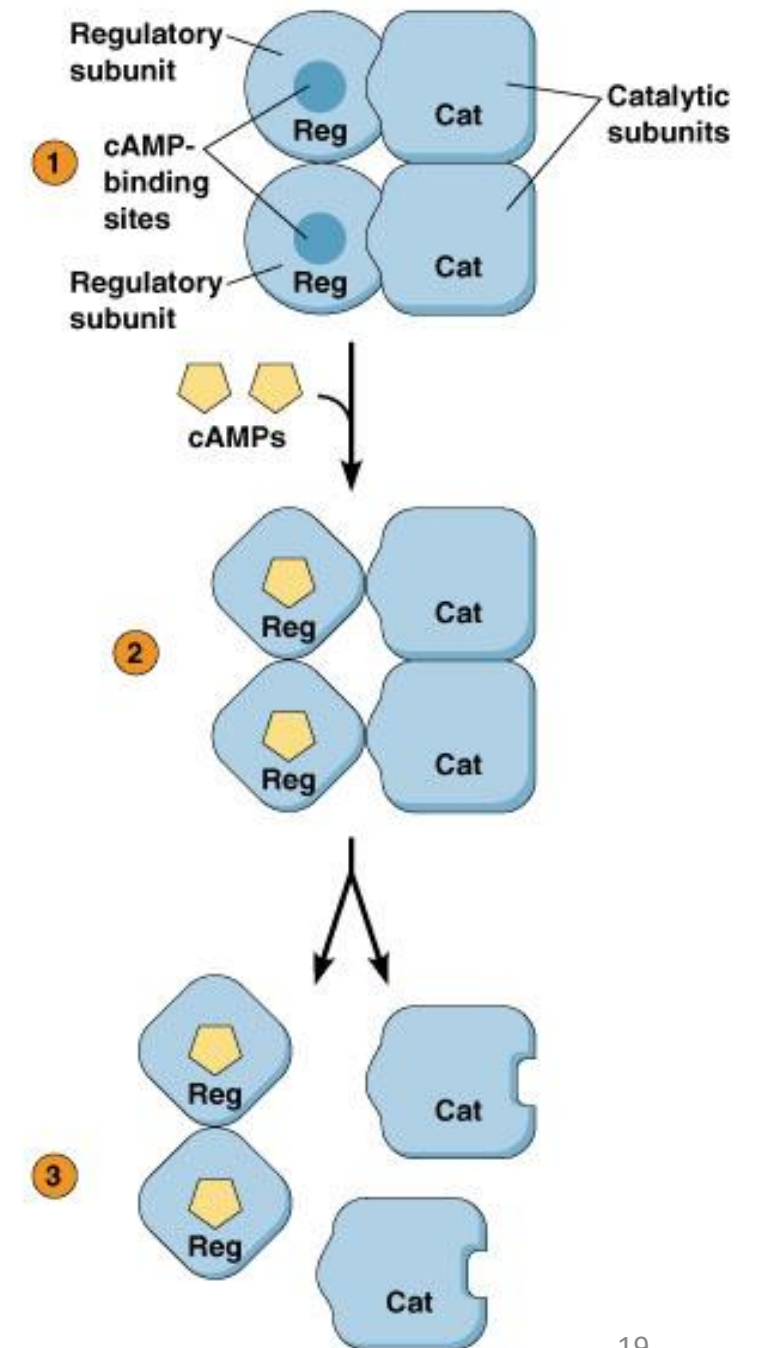
4. **Adenylate cyclase** catalyzes the conversion of ATP to **cAMP** (second messenger)

cAMP as a second messenger

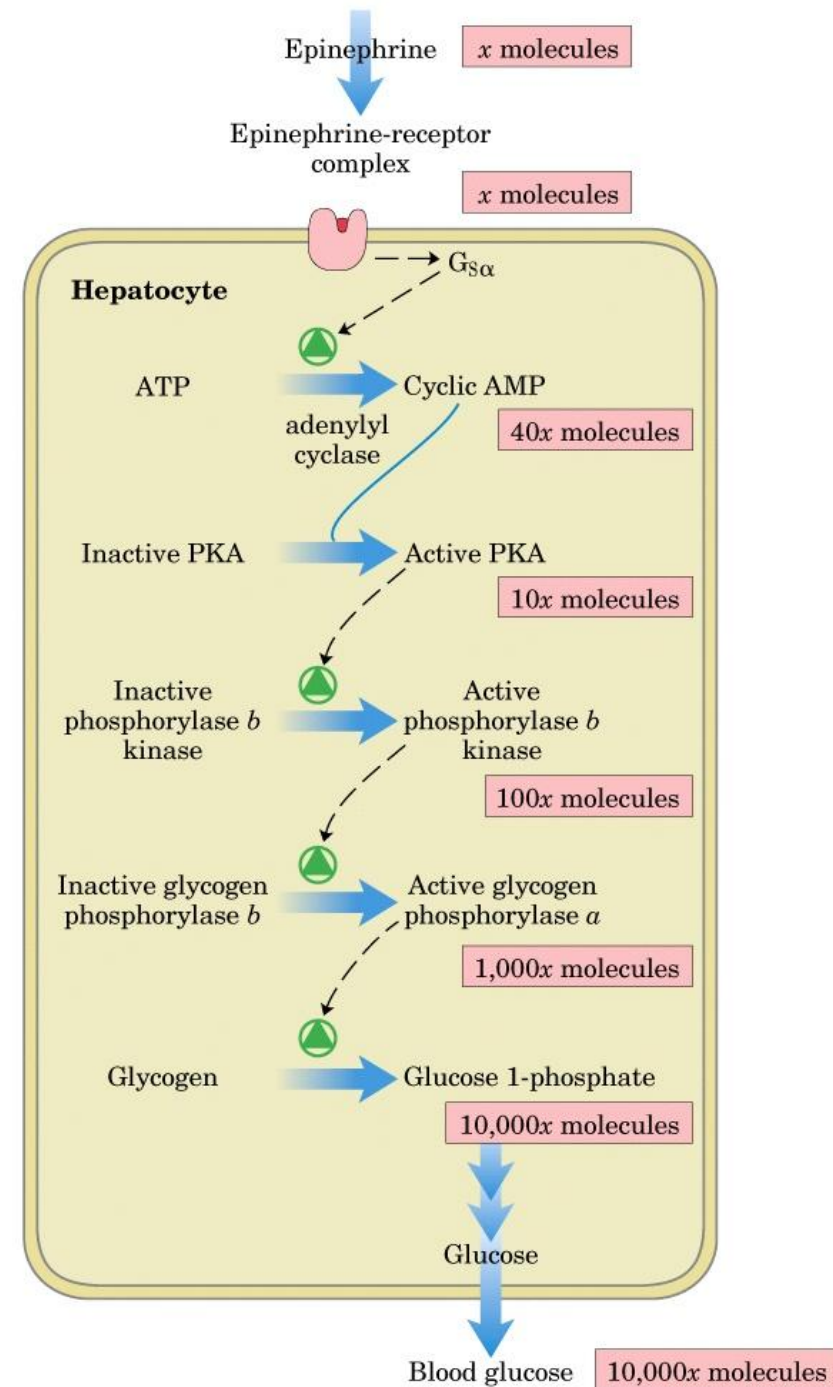
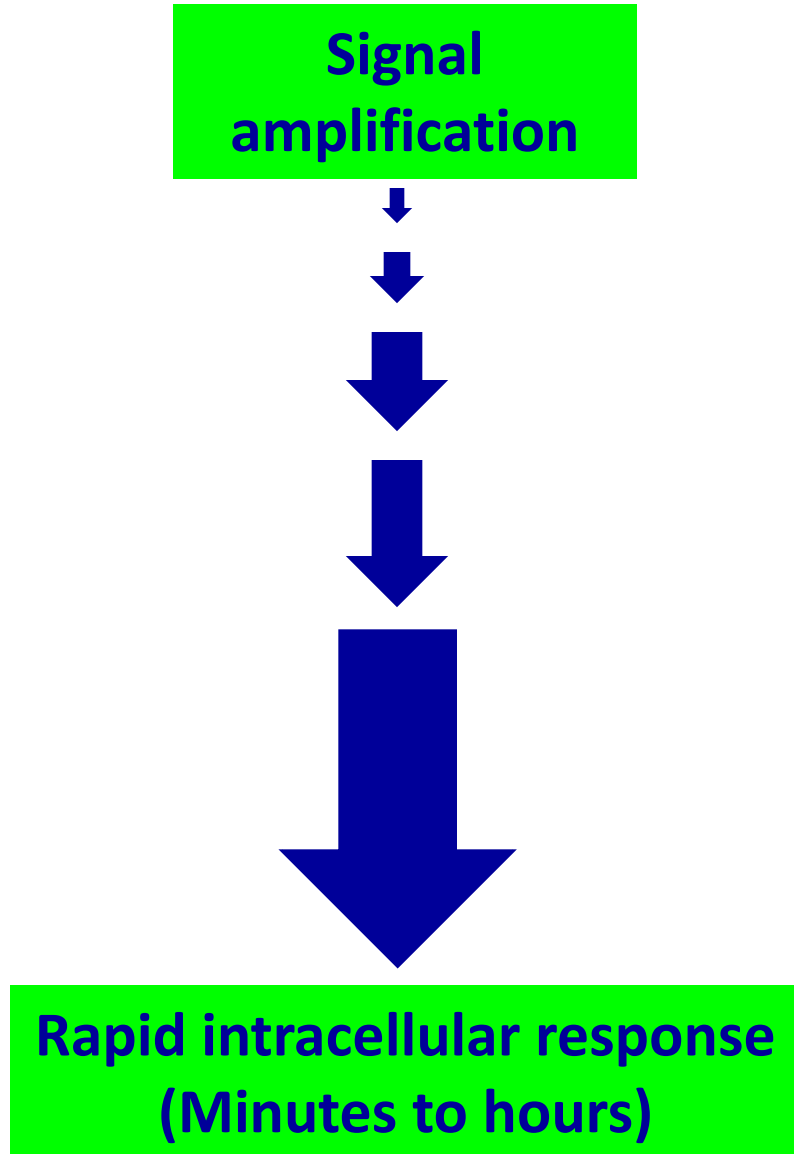
5. cAMP binds to regulatory subunits of **protein kinase A**; Catalytic kinase subunits are active

6. Catalytic **kinase subunits** transfer phosphate from **ATP** to **Ser** or **Thr** residues of protein substrates (enzymes) - **Covalent modification** of enzymes

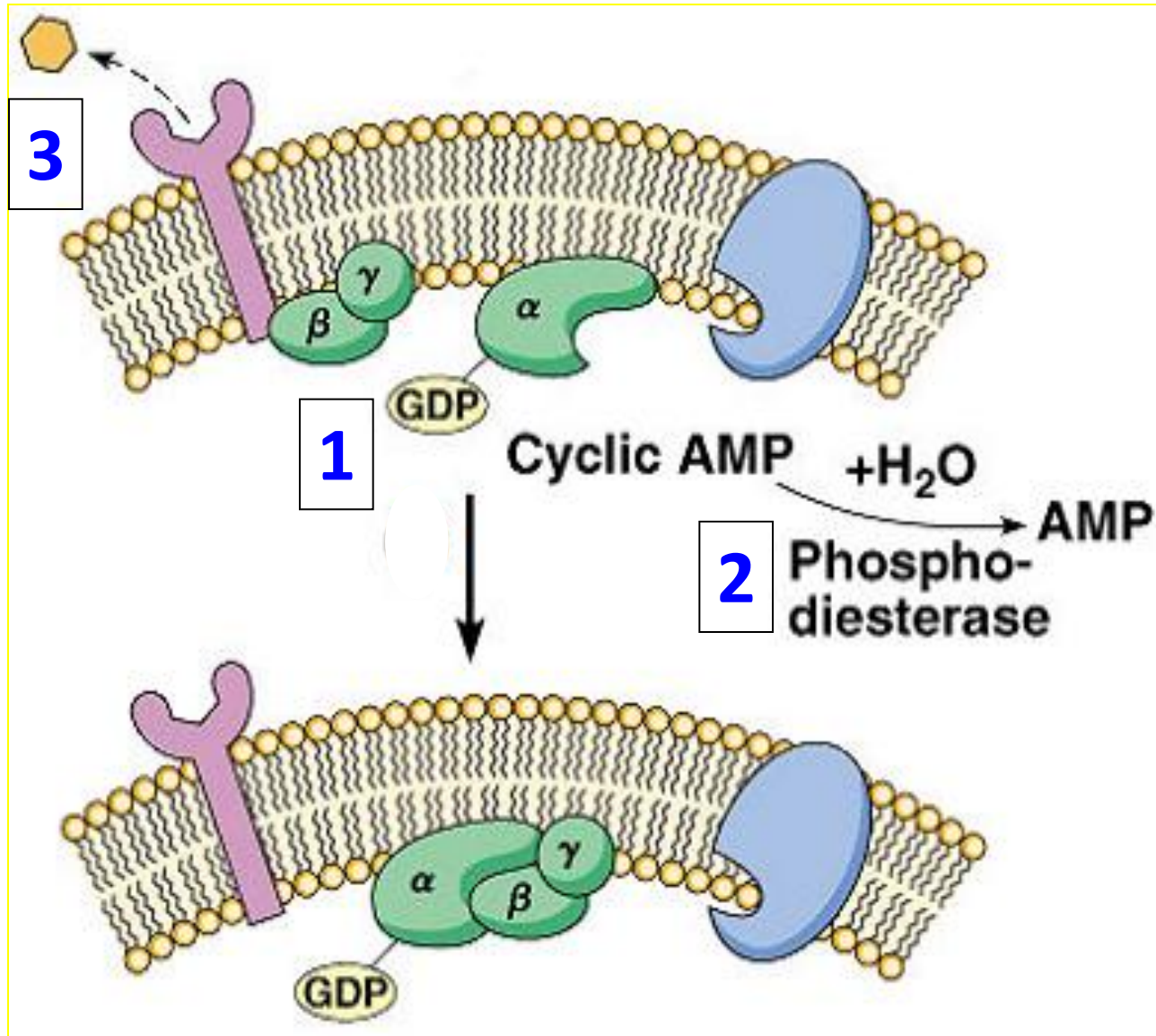
7. **Phosphorylation** of enzymes can activate or inhibit enzymes → Regulation metabolic pathways



Adenylate cyclase system

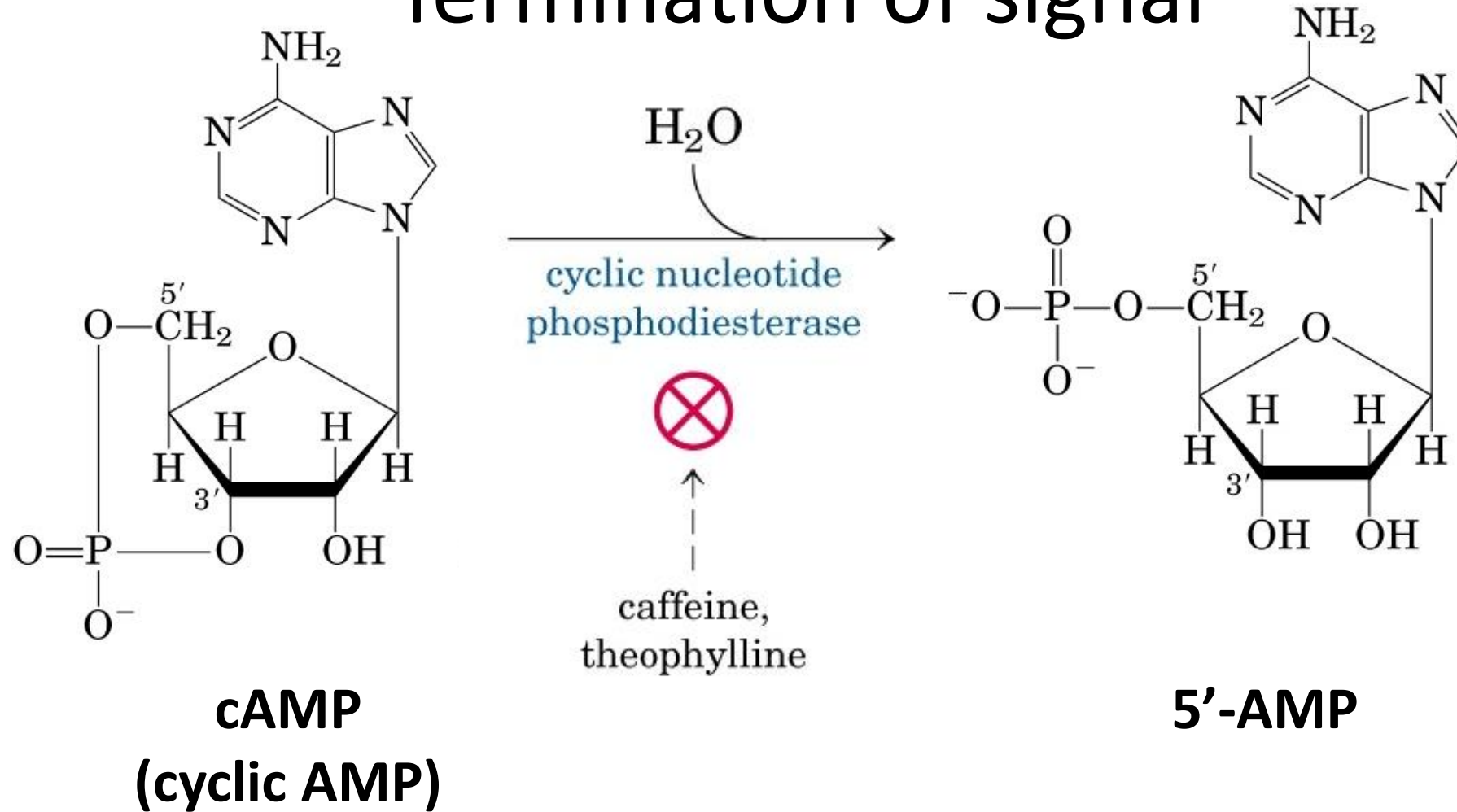


Termination of signal



1. Intrinsic **GTP** → **GDP + Pi** **GTPase** activity of **G-α subunit** (Intrinsic timer).
2. **Phosphodiesterase** enzyme hydrolyzes cAMP to AMP
3. Hormone dissociates from receptor

Termination of signal



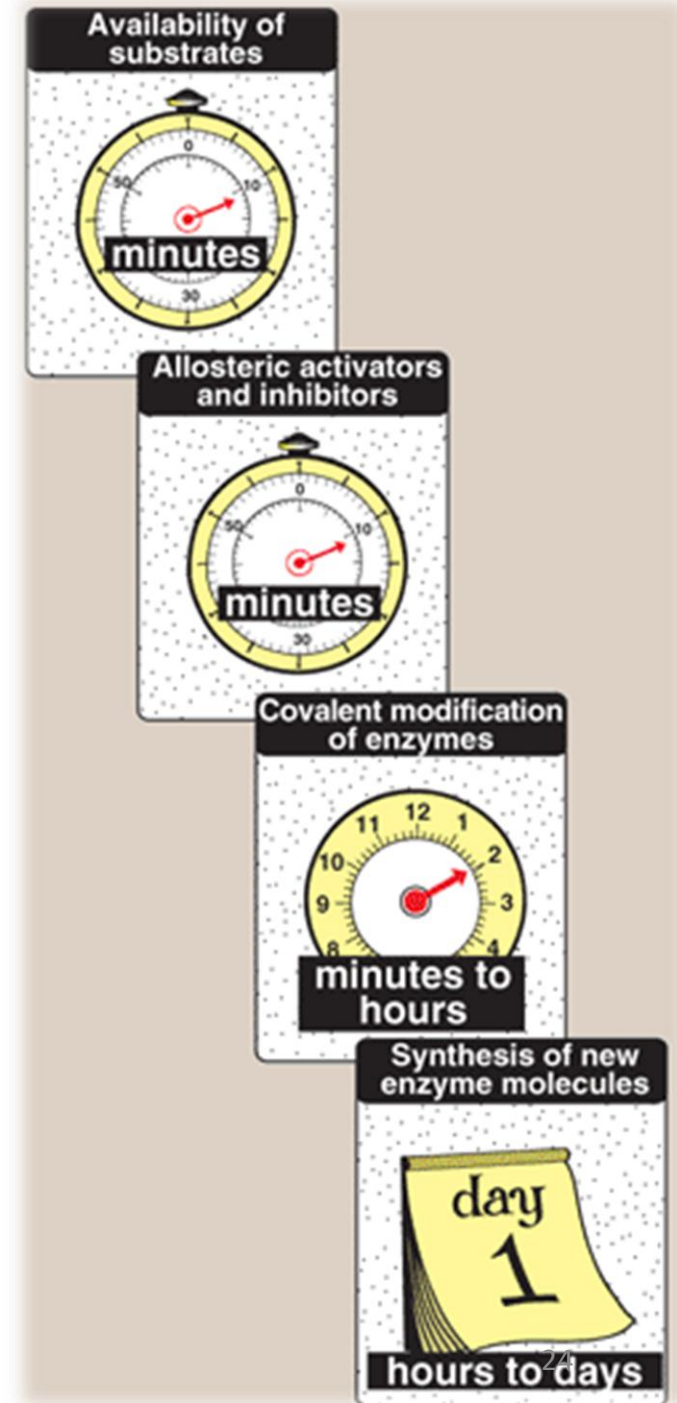
Caffeine and theophylline inhibit cyclic nucleotide phosphodiesterase and prolong effects of cAMP

Adenylate cyclase system - Summary

- H-R complex activates $G\alpha_s$
- $G\alpha_s$ -GTP activates **adenylate cyclase**
- Second messenger is **cAMP**
- cAMP activates **protein kinase A (PKA)**
- **Protein kinase A** phosphorylates **target proteins/enzymes**
- **Termination:**
 - **Intrinsic GTPase** of $G\alpha$
 - **Phosphodiesterase** hydrolyzes cAMP to 5'-AMP

Enzyme regulatory mechanisms

- Availability of substrates (minutes)
- Allosteric activation /inhibition of enzymes (minutes)
- Covalent modification of enzymes (minutes to hours)
 - mediated via Protein Kinase A
- Induction (up-regulation) or repression (down-regulation) of enzyme synthesis (hours to days)

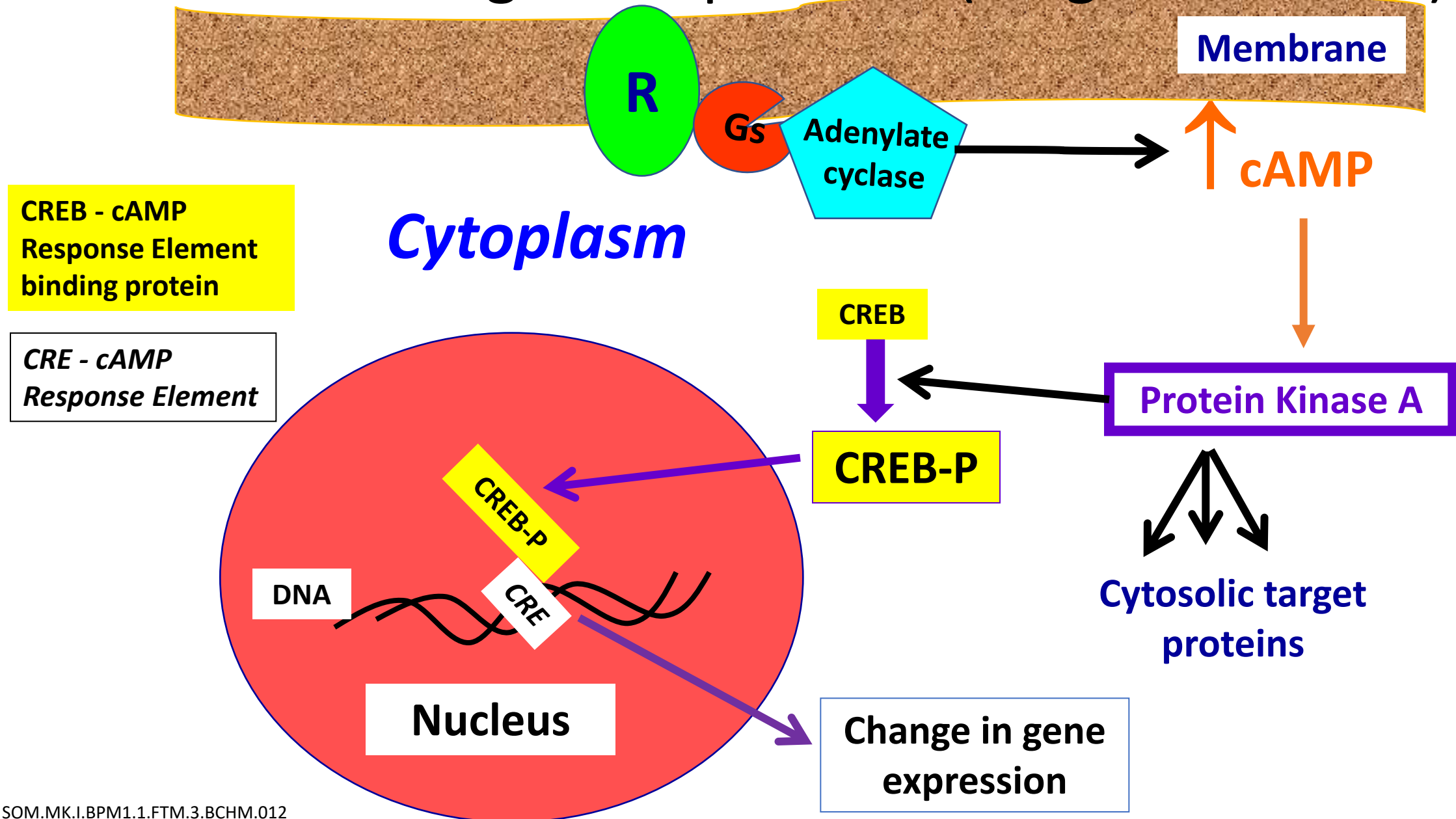


cAMP exerts effects via activation of PKA: Rapid effect (seconds to minutes) - covalent modification of pre-existing enzymes

Glycogen synthase Phosphorylase <i>b</i> kinase	Glycogen synthesis Glycogen breakdown
Pyruvate kinase (rat liver)	Glycolysis
Pyruvate dehydrogenase complex (type L)	Pyruvate to acetyl-CoA
Hormone-sensitive lipase	Triacylglycerol mobilization and fatty acid
Phosphofructokinase-2/fructose 2,6-bisphosphatase	Glycolysis/gluconeogenesis
Tyrosine hydroxylase	Synthesis of L-DOPA, dopamine, norepinephrine, and epinephrine
Histone H1	DNA condensation
Histone H2B	DNA condensation
Cardiac phospholamban (a cardiac pump regulator)	Regulation of intracellular $[Ca^{2+}]$
Protein phosphatase-1 inhibitor-1	Regulation of protein dephosphorylation
CREB	cAMP regulation of gene expression

**cAMP/ PKA also exerts effects via gene regulation through CREB:
Response Relatively slow – hours to days**

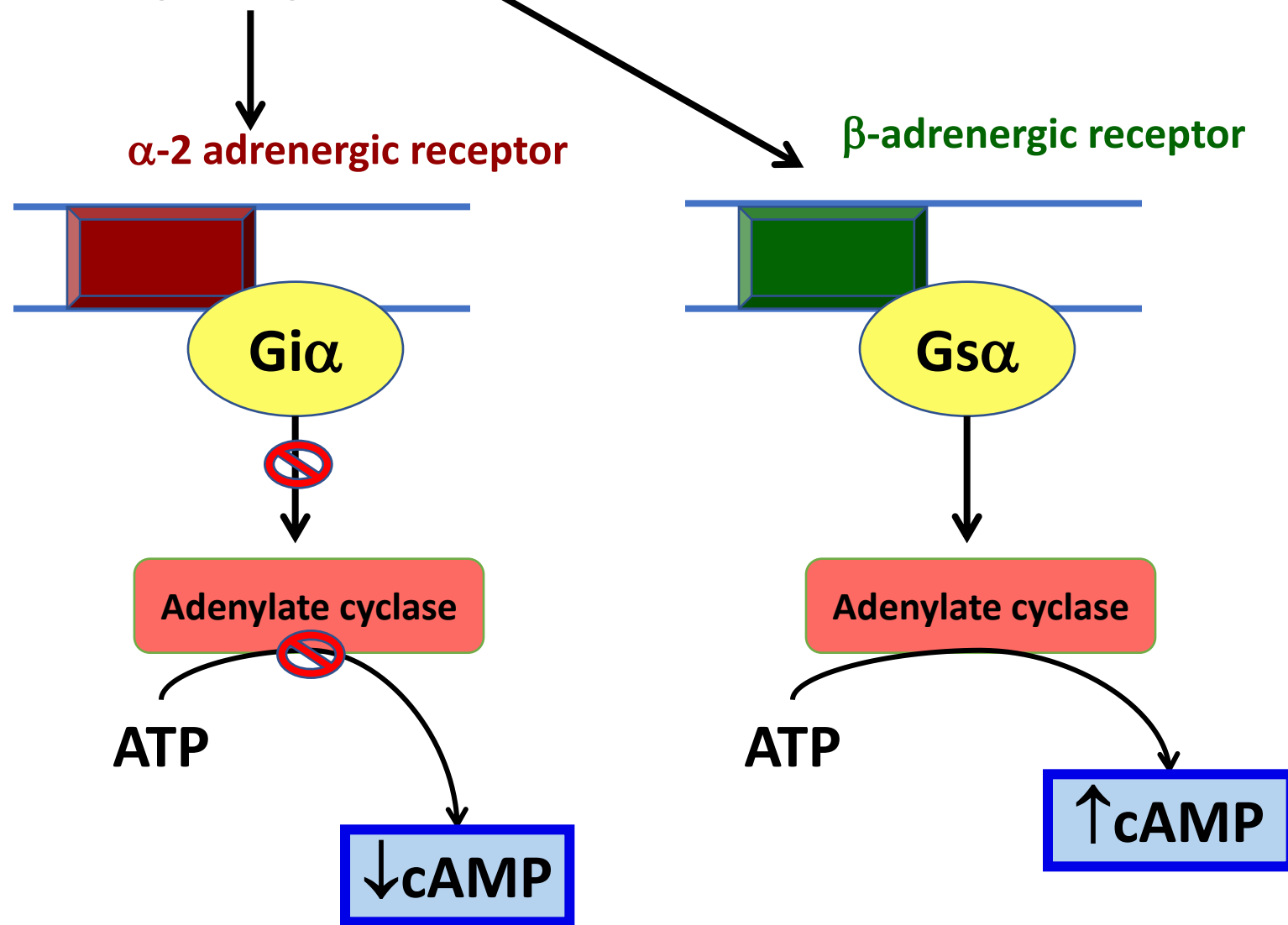
cAMP: Effect on gene expression (long term effect)



α 2-Adrenergic receptors

- Epinephrine binds to α 2-adrenergic receptors activates **G α**
- **Inhibits** Adenylate cyclase
- Reduces intracellular cAMP levels
- Compare α 2 and β

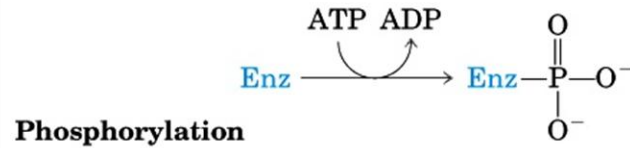
Epinephrine or Norepinephrine



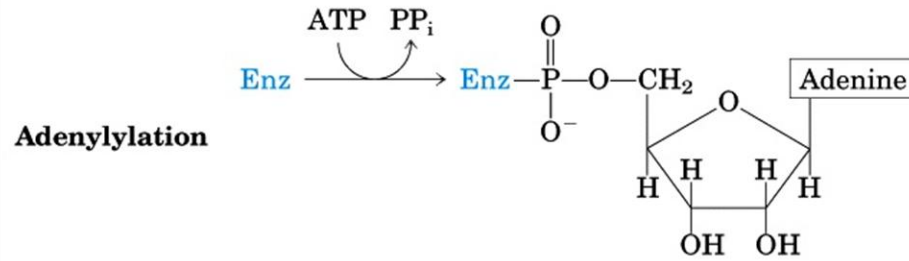
Modification of G protein

Covalent modification

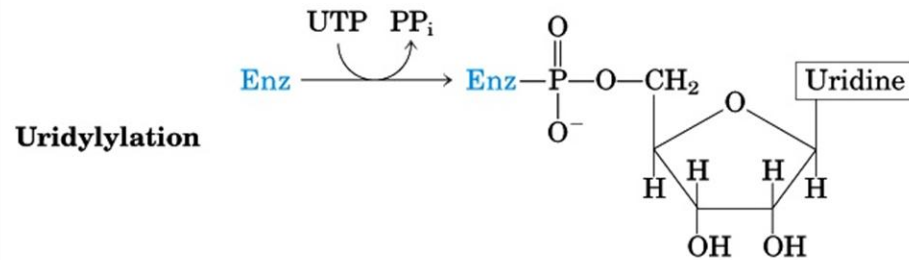
Amino acid residues known to accept covalent modification



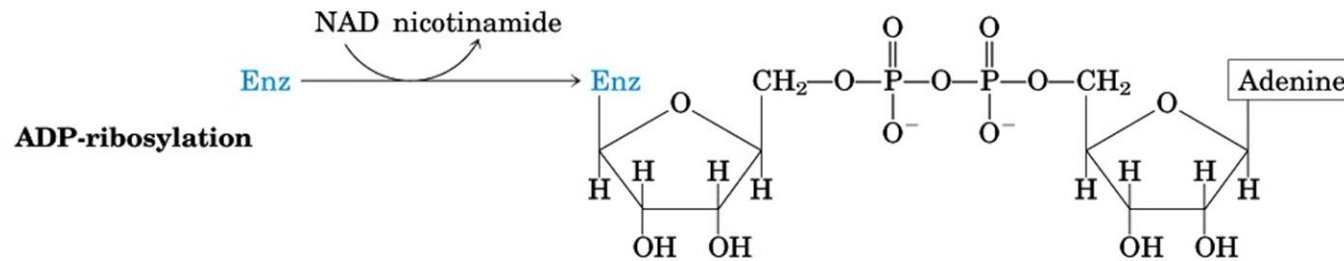
Tyr, Ser, Thr, His



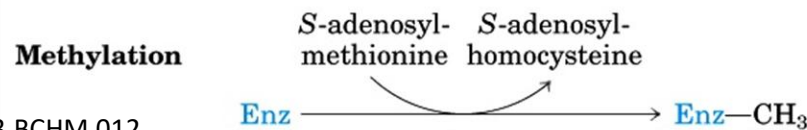
Tyr



Tyr



Arg, Gln, Cys, diphthamide (a modified His)



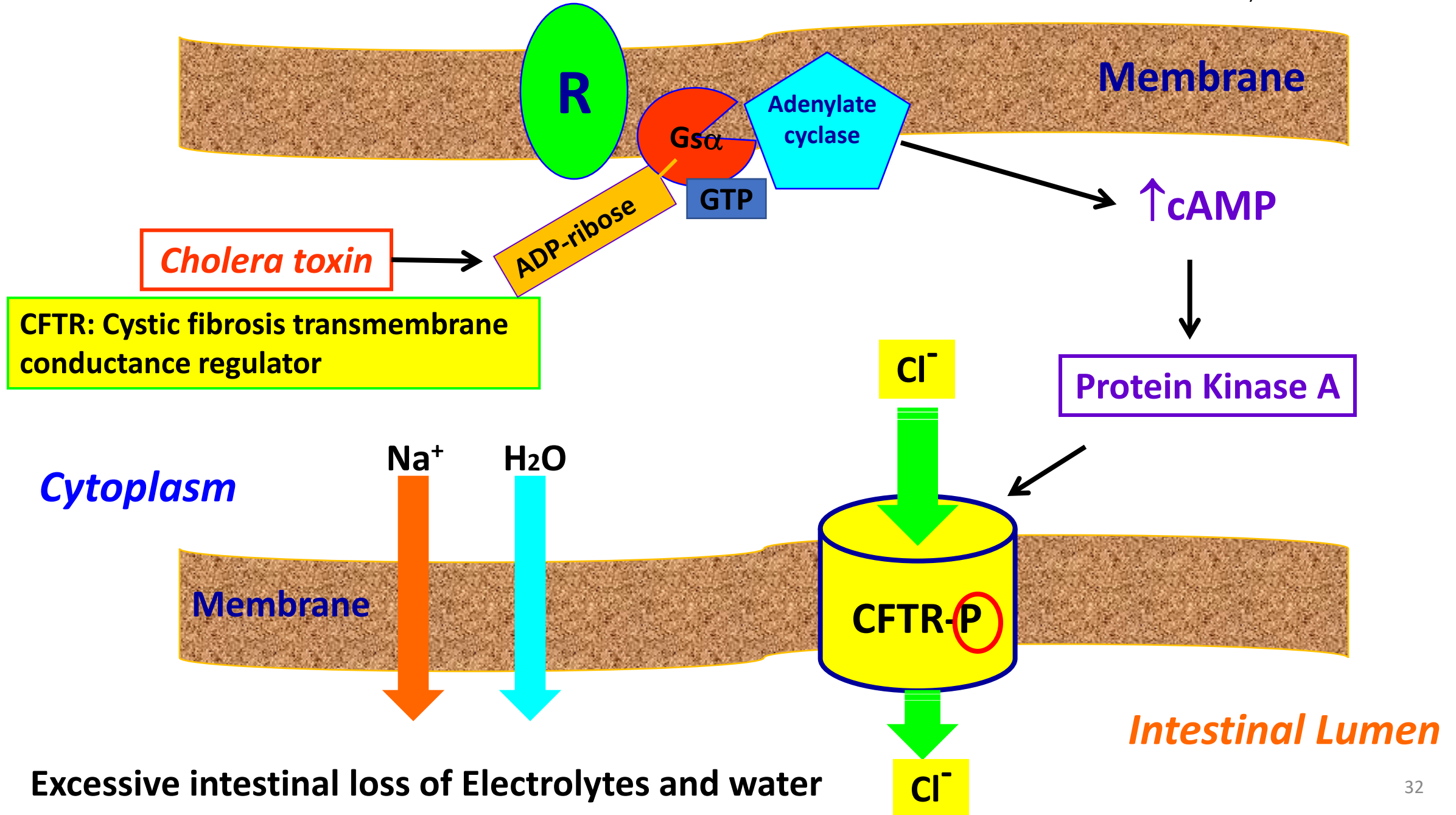
Glu

Regulation of enzyme activity

Covalent Modification of G α subunits- Bacterial toxins

- **ADP-ribosylation** (Addition of ADP-ribose) → **Inactivation** of proteins
- **Cholera toxin** ADP-ribosylates **G α s** → **Continuously active G α s**
 - Intrinsic GTPase activity of **G α s** destroyed, G α s locked in GTP-bound state
 - **Increased cAMP** in intestinal mucosal cells
 - Opening of CFTR → Loss of electrolytes and water → Severe watery diarrhea
- **Pertussis toxin** ADP-ribosylates **G α i** → **Continuously inactive G α i**
 - Increased cAMP in respiratory-tract cells
 - Respiratory distress, and whooping cough
- Diphtheria toxin ADP-ribosylates eEF-2 (translation)
- *Some pathogenic E. coli toxin also ADP ribosylates Gs α*

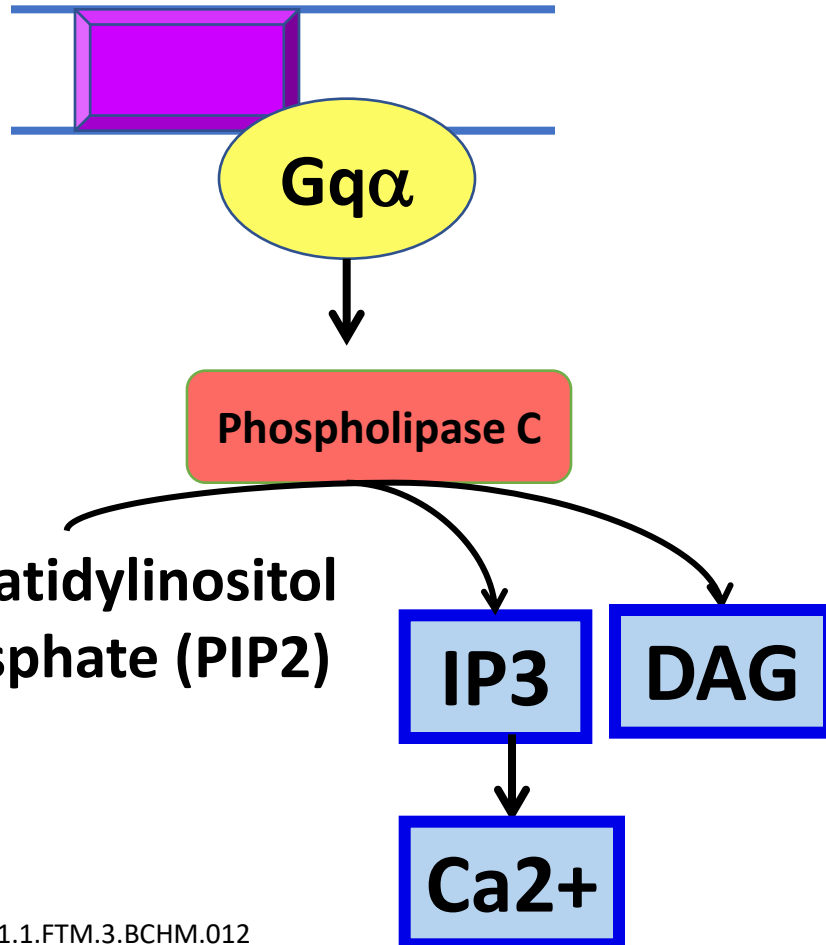
Cholera toxin: Mechanism of action



**Epinephrine or
Norepinephrine**

**α -1 adrenergic receptor:
second messengers**

α -1 adrenergic receptor



α -1 adrenergic receptor: second messengers

Stimulus: Epinephrine / Norepinephrine



Receptor: α 1-adrenergic receptor

- Activates **Gq α**
- Activates **Phospholipase C**
- Forms Second messengers
 - Inositol Tris-phosphate (IP3)
 - Calcium
 - Diacylglycerol (DAG)
- Activates **Protein kinase C**

Action of phospholipase C on Phosphatidylinositol-4,5-bis-phosphate (**PIP₂**)

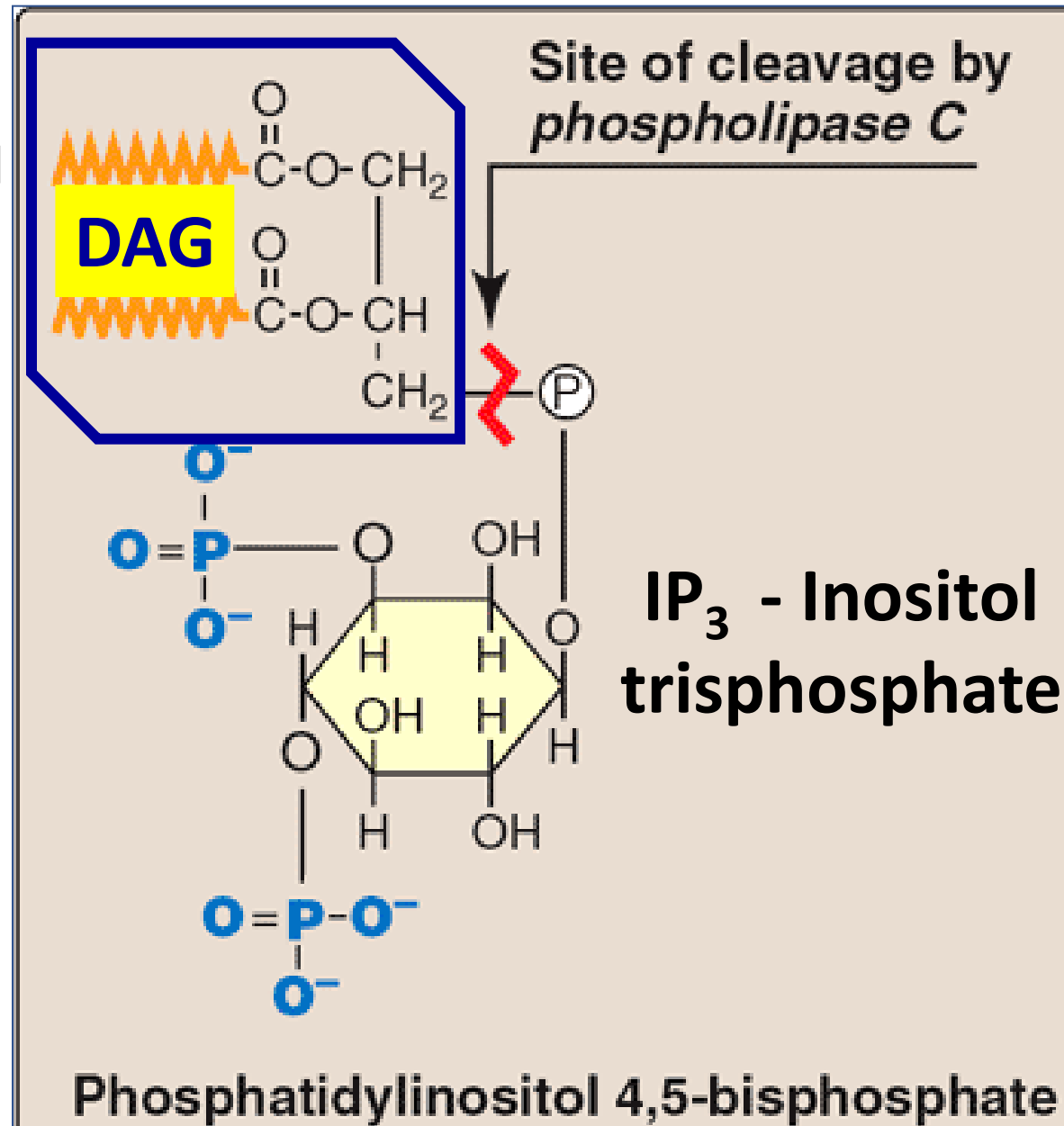
Fatty acid

Fatty acid

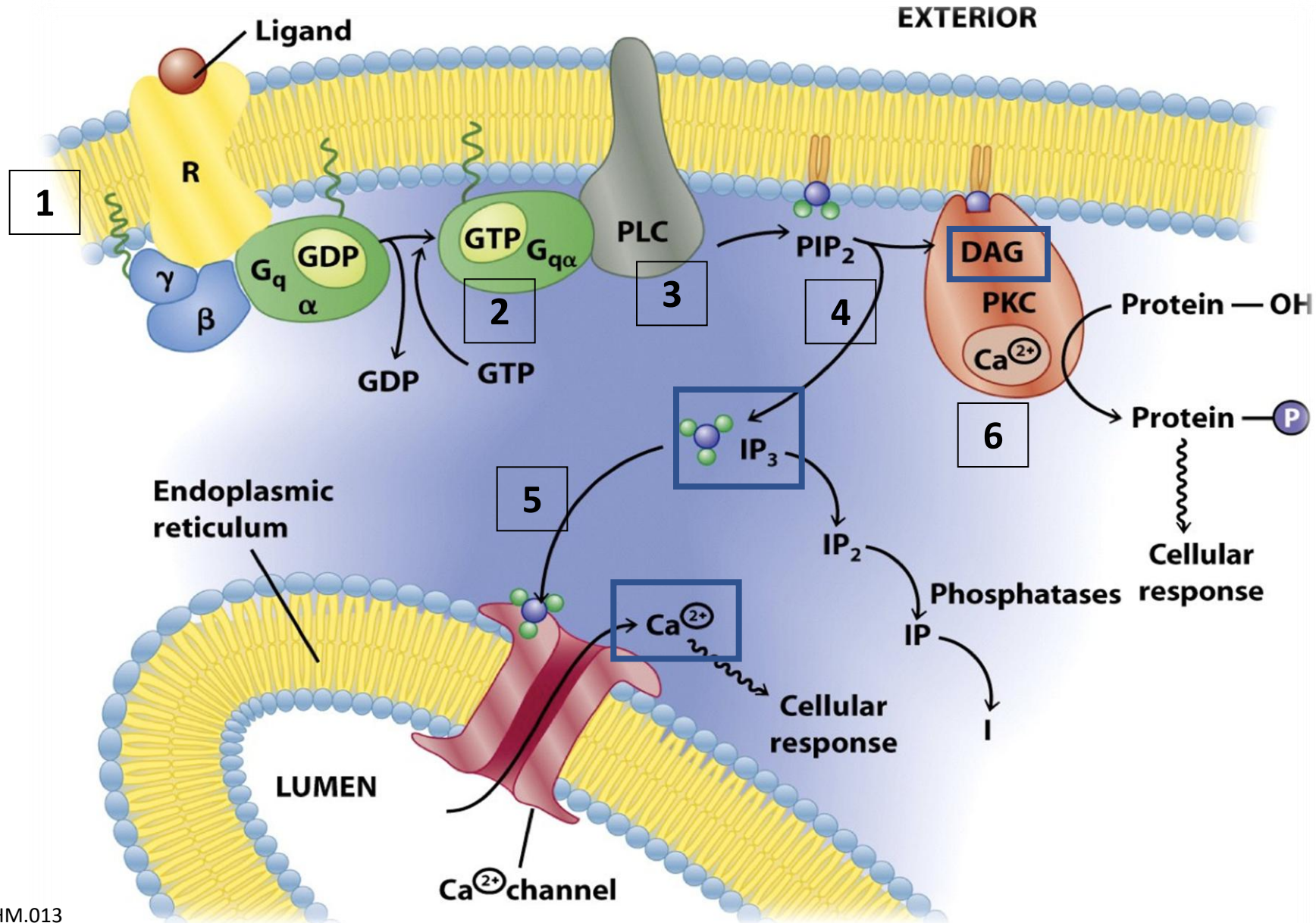
DAG

Site of cleavage by
phospholipase C

Phospholipase C
cleaves **PIP₂** to generate
IP₃ and **DAG**



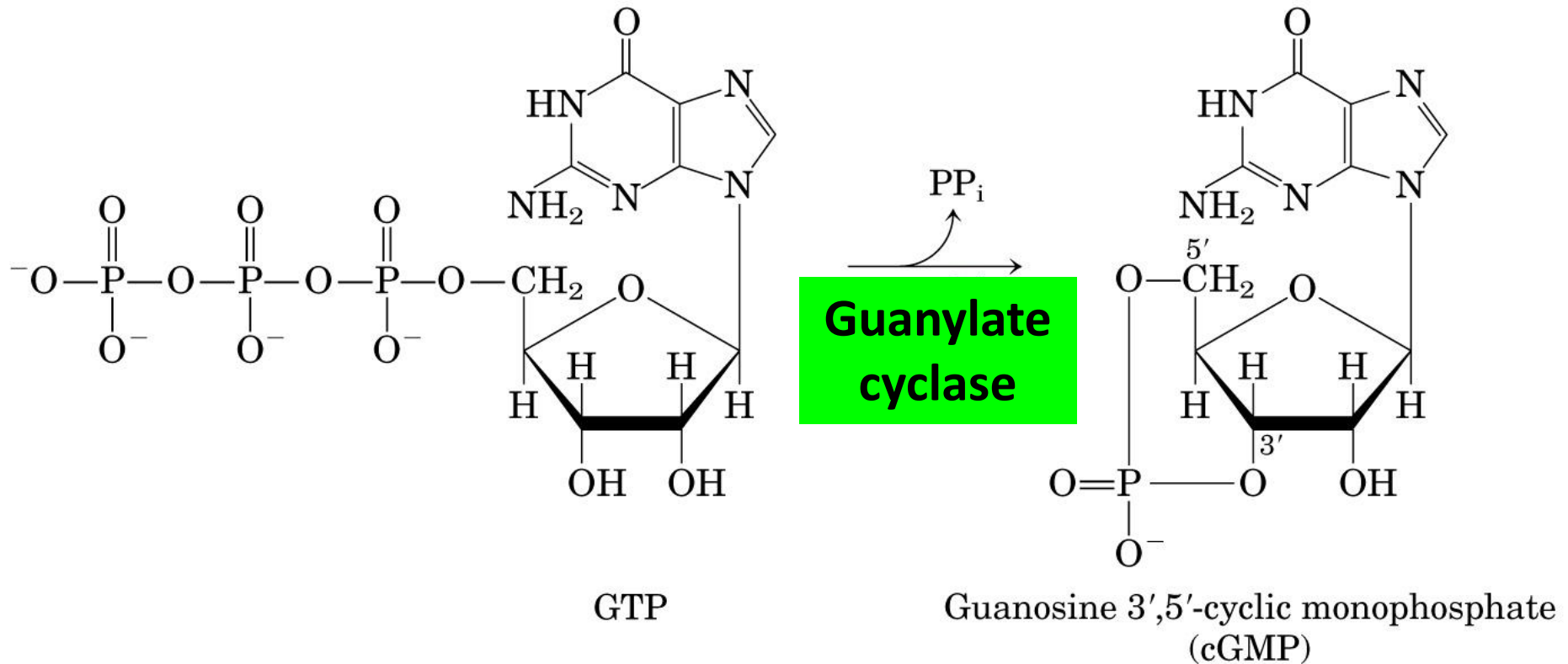
Intracellular events: Phosphoinositide system



Phosphoinositide system

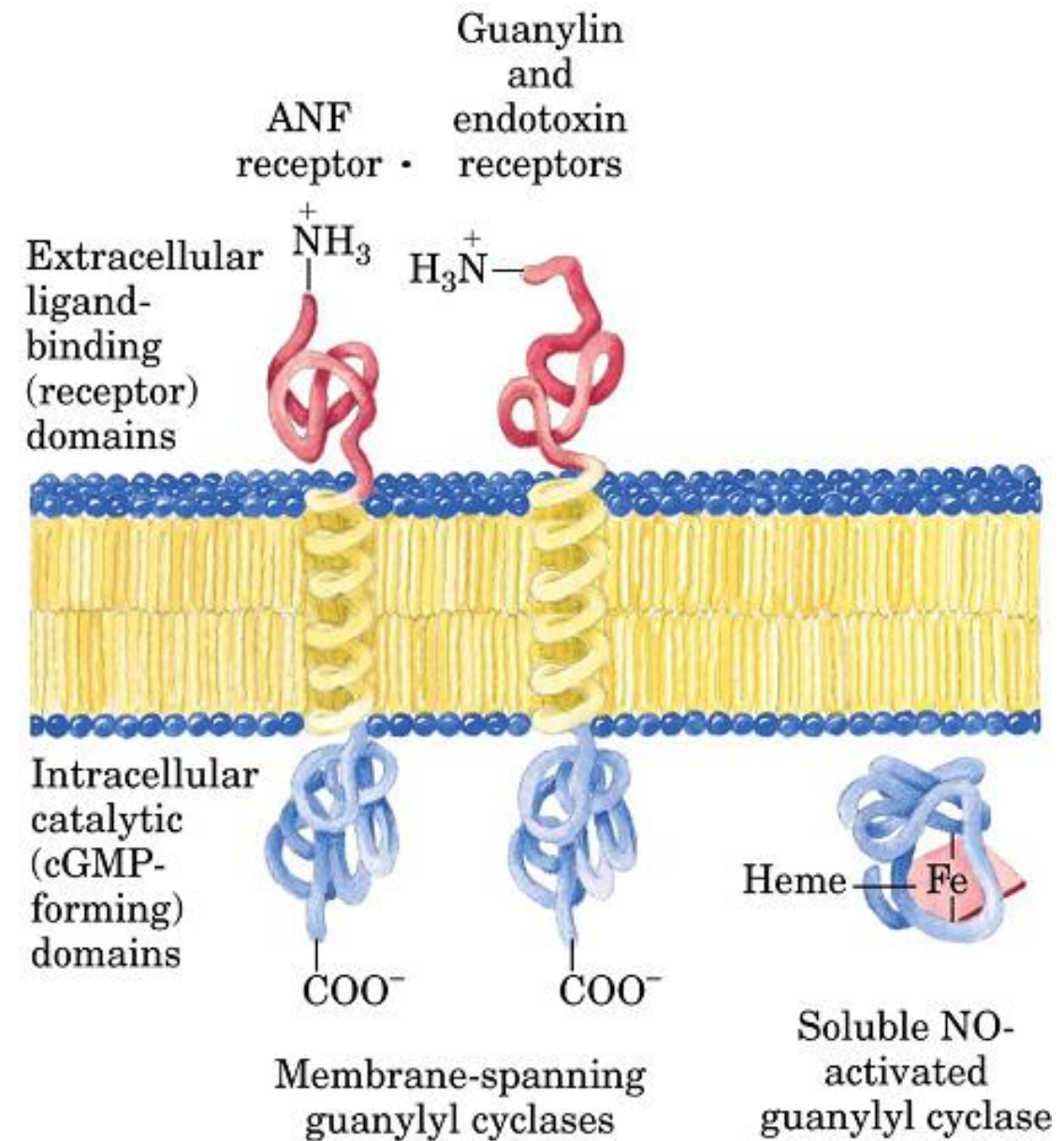
- Second messengers are **IP₃**, **DAG** and **Ca²⁺**
 1. Epinephrine binds to **α1-adrenergic** receptor
 2. Activates **Gqα** which binds to **GTP (in exchange for GDP)**
 3. **Gqα-GTP** activates **phospholipase C** (PLC)
 4. **PLC** cleaves **PIP₂** to **IP₃** and **DAG**
 5. **IP₃** causes **Ca²⁺** release from endoplasmic reticulum
 6. **DAG** and **Ca²⁺** activate **Protein kinase C** for cellular responses
 - **PKC** requires **DAG**, **Ca²⁺** and **phospholipids** (membrane)
- **Cholinergic muscarinic receptors** activate phospholipase C isoform to form **IP₃**, **DAG** and **Ca²⁺**

cGMP: Second messenger in smooth muscle relaxation, platelet aggregation and in visual system



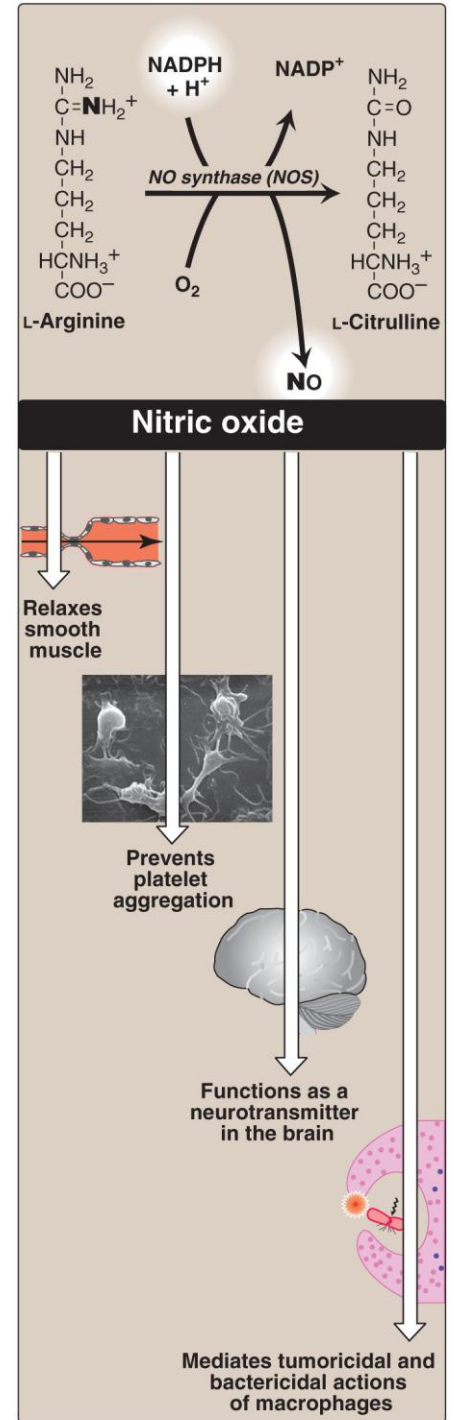
Guanylate cyclase

- Two forms
 - Membrane-bound **guanylate cyclase** (Binds to signals outside cell and produce cGMP inside cell)
 - Activated by atrial natriuretic peptide)
 - **Cytosolic guanylate cyclase (activated by NO)**
- G-protein is NOT involved
- Compare to adenylate cyclase and cAMP system mediated by G-protein



Nitric Oxide (NO)

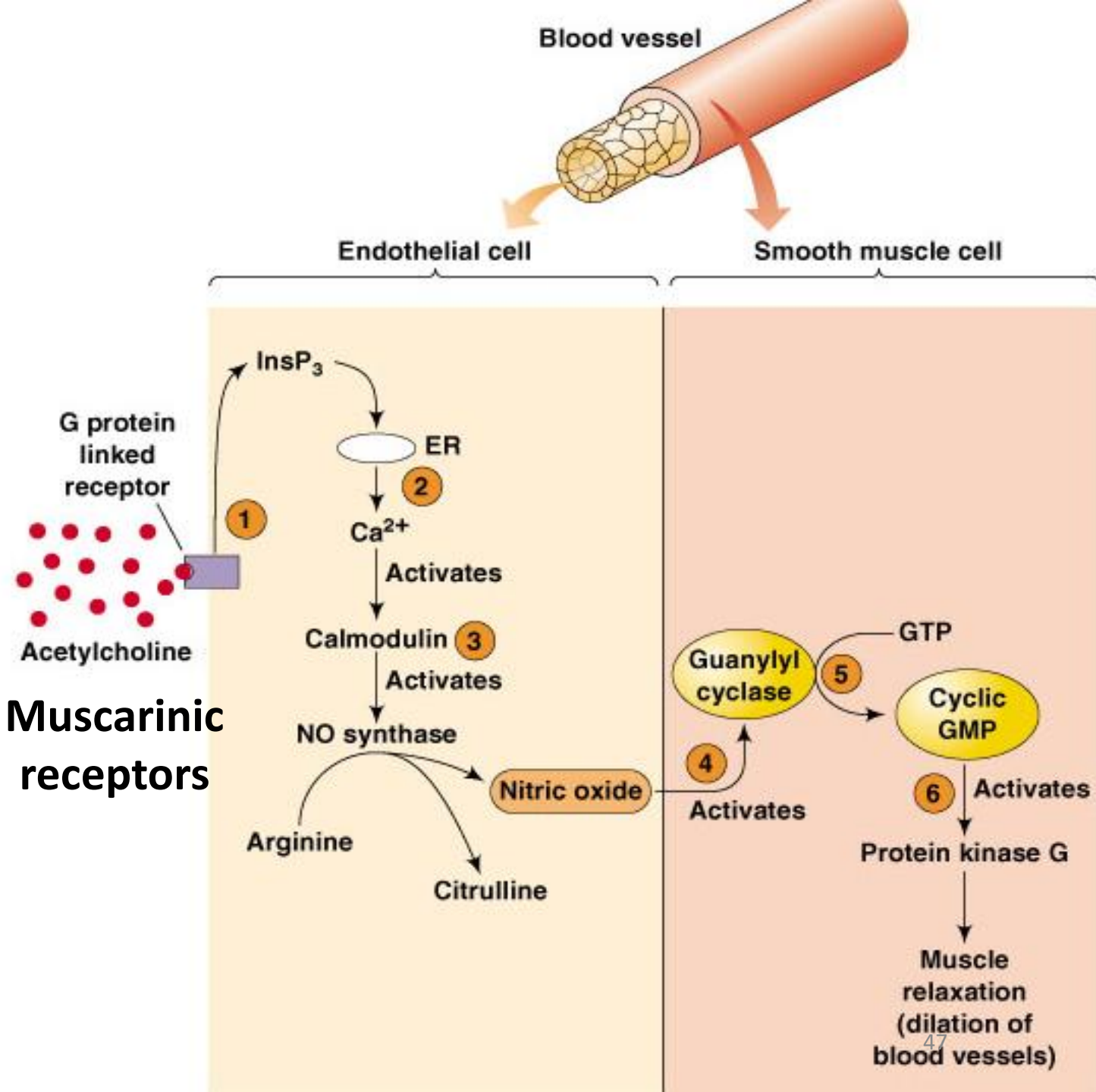
- NO (gas) synthesized from **Arginine**
- NO (gas) synthesized in endothelial cells (capillaries) diffuses to vascular smooth muscle where it activates **cytosolic guanylate cyclase**
- **cGMP levels** increase within smooth muscle cells
- Activates **Protein kinase G** which phosphorylates various proteins
- **Vascular smooth muscle relaxation** by leading to phosphorylated myosin light chain kinase and **inhibits** it
- Less phosphorylation of myosin light chains (vasodilation)



G-protein coupled receptors,
and NO work together to
regulate vasodilation

Involvement of several
intracellular second
messengers to
regulate vasodilation
(IP₃, Ca²⁺, NO, cGMP)

Muscarinic receptors



NO and cGMP in smooth muscle relaxation (vasodilation)

1. Activation of cholinergic (muscarinic) receptors in the endothelium by **acetylcholine**
2. Activation of **phospholipase C** (isoenzyme) results in release of **IP3** → and increases intracellular calcium
3. Calcium binds to calmodulin → activates NO synthase, producing **NO** in endothelial cells
4. NO (gas) **diffuses** to adjacent smooth muscle cells and activates soluble **guanylate cyclase**
5. Increases **cGMP**
6. Activation of **PKG** and subsequent muscle relaxation

Nitric Oxide (NO)

- Endogenous vasodilator (aka Endothelium-derived relaxing factor)
- Drugs like **nitroglycerin and nitrates (Antianginal/ antihypertensive)**
 - Nitroglycerin/ nitrates metabolized to NO
 - Angina (chest pain) is due to less myocardial oxygen supply
 - NO causes vasodilation of coronary blood vessels; systemic blood vessels and reduce cardiac load
 - Used as vasodilators

Summary

Hormone	Signal transduction pathway
Alpha1-adrenergic receptor	Activates Gq-alpha → Phospholipase C → DAG and IP3 → IP3 increases Ca ²⁺ → DAG and Ca ²⁺ Activate Protein kinase C
Alpha2-adrenergic receptor	Activates Gi-alpha → Inhibits adenylate cyclase → Decreases cAMP
Beta adrenergic receptor	Activates Gs-alpha → Stimulates adenylate cyclase → Increases cAMP → Activates protein kinase A → Phosphorylation of enzymes; CREB – Gene expression Phosphodiesterase cleaves cAMP
Glucagon receptor	Activates Gs-alpha → Stimulates adenylate cyclase → Increases cAMP → Activates protein kinase A → Phosphorylation of enzymes; CREB – Gene expression
Nicotinic ACh receptor	Binds Gated ion channel → Opens Na ⁺ channel
Insulin receptor	Activates tyrosine kinase activity → Tyrosine phosphorylation of IRS → Activation of PI3 Kinase
Steroid-thyroid family hormones	Bind intracellular receptors → Form Hormone-receptor complex → Bind to HRE → Induce or repress gene expression
Nitric oxide	Activates guanylate cyclase → Increases cGMP → Activates Protein kinase G → Vasodilation